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Cosmology with the Wide-Field Infrared Survey Telescope - Synergies with the Large Synoptic Survey Telescope

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ABSTRACT

We explore synergies between the space-based Wide-Field Infrared Survey Telescope (WFIRST) and the ground-based Large Synoptic Survey Telescope (LSST). In particular, we consider a scenario where the currently envisioned survey strategy for WFIRST's High Latitude Survey (HLS), i.e. 2000 deg^2 in four narrow photometric bands is altered in favor of a strategy that combines rapid coverage of the LSST area (to full LSST depth) in one band.

We find that already a 5-month WFIRST survey in the W-band can cover the full LSST survey area providing high-resolution imaging for >95% of the LSST Year 10 weak lensing galaxy sample. We explore a second, more ambitious scenario where WFIRST spends 1.5 years covering the LSST area. For this second scenario we quantify the constraining power on dark energy equation of state parameters from a joint weak lensing and galaxy clustering analysis and we compare it to an LSST-only Year 10 survey and to the standard HLS survey.

Our survey simulations are based on the WFIRST exposure time calculator and redshift distributions from the CANDELS catalog. Our statistical uncertainties account for higher-order correlations of the density field, and we include a wide range of systematic effects, such as uncertainties in shape and redshift measurements, and modeling uncertainties of astrophysical systematics, such as galaxy bias, intrinsic galaxy alignment, and baryonic physics. We find a significant increase in constraining power for the joint LSST+WFIRST wide survey compared to LSST Y10 (FoM$_{\rm Wwide}$= 2.6 FoM$_{\rm LSST}$) and compared to WFIRST HLS (FoM$_{\rm Wwide}$= 4.4 FoM$_{\rm HLS}$).

Key words: cosmological parameters – theory –large-scale structure of the Universe

1 INTRODUCTION

[Chris: confirm specs in Table 1]

Observing the the large-scale structure in our Universe provides a wealth of cosmological information and allows us to constrain fundamental physics questions such as the nature of dark energy, the mass and number of species of neutrinos, possible modifications to general relativity as a function of scale or environment, or the nature of dark matter interactions.

Early results from the Dark Energy Survey (DES[3], Krause et al. 2017; Troxel et al. 2018; Abbott et al. 2018, 2019), the

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[3] www.darkenergysurvey.org/

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Table 1. Technical specifications characterizing the photometric data sets from LSST and WFIRST.

WFIRST Imaging Capabilities [1]							
Telescope Aperture		**Field of View**		**Pixel Scale**		**Wavelength Range**	
2.4 meter		**0.28 sq deg**		**0.11 arcsec**		**0.5-2.0 μm**	
Filters	R062	Z087	Y106	J129	H158	F184	W146
Wavelength (μm)	0.48 - 0.76	0.76 - 0.98	0.93 - 1.19	1.13 - 1.45	1.38 - 1.77	1.68 - 2.00	0.93 -2.00
Sensitivity (5 σ AB mag in 1h)	28.5	28.02	27.95	27.87	27.81	27.32	28.33

LSST Imaging Capabilities [2]						
Telescope Aperture		**Field of View**		**Pixel Scale**		**Wavelength Range**
6.67 meter (effective)		**9.62 sq deg**		**0.2 arcsec**		**0.32 - 1.0 μm**
Filters	u	g	r	i	z	Y
Wavelength (μm)	0.32 - 0.40	0.40 - 0.55	0.55 - 0.69	0.69 - 0.82	0.82 - 0.92	0.92 - 1.0
Sensitivity (5 σ AB mag in 1h)						

Kilo-Degree Survey (KiDS[4], van Uitert et al. 2018; Hildebrandt et al. 2018; Kuijken et al. 2019), and the Hyper Suprime Cam (HSC[5], Hikage et al. 2019; Hamana et al. 2019) have demonstrated the feasibility of complex (multi-probe) analyses from photometric data. Within the ΛCDM model, these surveys show a ∼2-σ tension/agreement with Cosmic Microwave Background (CMB) measurements from the Planck satellite, which sets the stage for exciting near future discoveries when the full data of DES, HSC, KiDS is analyzed.

This discovery potential will increase significantly in the coming decade when photometric data from Stage 4 surveys become available (see Weinberg et al. 2013, for a review). The Large Synoptic Survey Telescope (LSST[6], LSST Science Collaboration et al. 2009), Euclid[7] (Laureijs et al. 2011), the Spectro-Photometer for the History of the Universe, Epoch of Reionization, and Ices Explorer (SPHEREx[8], Doré et al. 2014), and the Wide Field Infrared Survey Telescope (WFIRST[9], Spergel et al. 2015) complement each other in terms of area, depth, wavelength, and resolution and will provide a wealth of cosmological data to be mined by the community. The concert of cosmological endeavors in the 2020s also spans several spectroscopic data sets (including from Euclid and WFIRST) and CMB observations, which are highly synergistic to the photometric information from the aforementioned surveys.

In this paper we focus on synergies between WFIRST and LSST, specifically we explore alterations to the WFIRST observing strategy that maximize said synergies.

The Large Synoptic Survey Telescope will start commissioning and science verification in 2021 and is scheduled to be fully operational in 2022. It will provide the cosmology community with an unprecedented amount of high quality data from the ground. LSST is best described as a wide-fast-deep-repeat survey that can cover 18,000 deg^2 every 3 nights in 6 visible bands and that will flood the community with ∼20TB of imaging data per day. LSST is limited by the atmosphere in terms of seeing and wavelength coverage. For dark energy and other science cases the issues of accurate redshift measurements and blended galaxies (Dawson et al. 2016) are potentially limiting systematics.

The Wide Field Infrared Survey Telescope is scheduled to launch mid 2020s and it is probably best described as a multi-purpose space observatory, with a 100 times larger field-of-view than the Hubble Space Telescope. WFIRST's versatile capabilities address science cases ranging from exoplanets to galaxy evolution to fundamental physics. WFIRST's High Latitude Survey (HLS) is designed to constrain dark energy evolution and deviations from GR with excellent control of systematics via space-quality imaging, photometry across 4 near-infrared (NIR) bands, a 0.28 deg^2 field of view with a 0.11 arcsec pixel scale, and a ∼600 resolution grism. The currently envisioned survey area of 2000 deg^2 is covered by WFIRST in 1.6 years both in the imaging and spectroscopic component enabling deep spectroscopic galaxy redshift measurements over the same survey area as the imaging survey.

Table 1 summarizes some of the main technical specifications of WFIRST and LSST. Most obvious is the complementarity in multi-wavelength information with LSST pushing towards the blue end in the visible and WFIRST overlapping towards the red end and extending the color information into the near-infrared. We note that while the current HLS survey design plans on using the YJHF bands, there are several other filters available, in particular the W-band ranging from 0.93-2.0 microns.

In this paper we comment on several scenarios where a single WFIRST band is used for a rapid, deep coverage of the LSST area. The combination of color information down to 320nm from the ground, and high-resolution infrared coverage from space over 18,000 deg^2 can potentially unlock a new level of precision for the core science cases of both experiments, if and only if, systematics control is maintained at the new level. Throughout the paper we refer to this single band concept as *WFIRST wide* and we will contrast it with the current 4 NIR-band 2000 deg^2 design, which we refer to as *WFIRST HLS*.

[4] http://www.astro-wise.org/projects/KIDS/
[5] http://www.naoj.org/Projects/HSC/HSCProject.html
[6] https://www.lsst.org/
[7] https://sci.esa.int/web/euclid
[8] http://spherex.caltech.edu/
[9] https://wfirst.gsfc.nasa.gov/

2 MULTI-PROBE ANALYSIS BASICS - COMBINING WEAK LENSING AND GALAXY CLUSTERING

[All: Read, comment, suggest refs, especially on forecasting philosophy.
Elisabeth, Xiao: confirm equations matching cosmolike implementation]

2.1 Forecasting Philosophy

A large variety of cosmological probes can be extracted from wide-field imaging and spectroscopic WFIRST observations, such as various forms of weak lensing (e.g., cosmic shear using second and higher order summary statistics, magnification, galaxy-galaxy lensing, cluster weak lensing, and shear peak statistics), galaxy clustering (spectroscopic 3D and projected 2D), galaxy cluster number counts, CMB-lensing, Baryon Acoustic Oscillations, and redshift space distortions.

Statements about the constraining power of different survey concepts are a function of the exact ingredients for the data vector, which does not just include the combination of probes but also the exact scales and redshifts that are included, the systematics considered, the parameterization and priors of said systematics, and the priors and parameters of the cosmological model of interest.

Our forecasting philosophy is driven by the intent to model a realistic WFIRST and LSST 3x2pt analysis (joint weak lensing and galaxy clustering). A realistic assessment of the constraining power requires accurately computing three main components of a likelihood analysis: 1) Statistical uncertainties, 2) systematic uncertainties, 3) robust inference technique.

Regarding the statistical uncertainties, we use analytic, non-Gaussian covariance matrices that include higher order moments of the density field and super-sample variance and that have been compared to numerical simulations (albeit only in less constraining cases). We also use realistic survey simulations (see Sect. 3) to quantify the number density and redshift distributions of source and lens sample and to compute a realistic survey area for a given scenario.

Regarding systematic uncertainties, we include observational uncertainties such as shear calibration uncertainties and photo-z errors for lens and source sample. We parameterize a variety of astrophysics uncertainties: Galaxy bias uncertainties are captured through a combination of scale cuts and and nuisance parameters, similarly we include uncertainties from intrinsic galaxy alignment and baryonic physics modeling again through a combination of nuisance parameters and scale cuts. The exact implementation is described in Sect. 4.

Regarding robust inference, we opted for simulated MCMC analyses instead of Fisher matrices in order to avoid numerical uncertainties in computing high-dimensional (56 dimensions spanning cosmological and nuisance parameters) derivatives of the observables and in order to capture posterior probabilities beyond the multi-variate Gaussian.

We employ the CosmoLike framework (Eifler et al. 2014; Krause & Eifler 2017) for the simulated likelihood analyses, however we do not use the most sophisticated modules to model the density power spectrum and the subsequent projected power spectra. For example, we do not use a Boltzmann code to compute the initial power spectrum nor do we include reduced shear, non-Limber, or magnification in the calculation of the summary statistics. We reason that since the data vector is computed with the exact same module, the per cent uncertainty arising from these approximations does not alter our results.

However, the fact that we are using the much faster, forecasting routines of CosmoLike, we speed up our simulated likelihood analysis by a factor of $\geqslant 10$, from 10 sec per MCMC step to ~ 0.7 secs, which allows for long, converged chains and enables a future in-depth survey scenario study that varies the baseline choices and priors presented in Table 2.

2.2 Modeling Details

In this section we summarize the computation of angular (cross) power spectra for the different probes closely following the notation in (Krause & Eifler 2017); a more detailed derivation can be found in Hu & Jain (2004). We use capital Roman subscripts to denote observables, $A, B \in \{\kappa, \delta_g\}$, where κ references lensing of source galaxies and δ_g is the density contrast of the lens galaxy sample.

We note the frequent confusion originating from the terms "lens sample" and "lensed galaxy sample" and clarify that we only use the position information of the lens sample. Specifically, the auto-correlation of positions of lens sample galaxies will later form the galaxy clustering part of the 3x2pt data vector, whereas the autocorrelation of lensed galaxy shapes, aka the cosmic shear part of the data vector, is computed using the source galaxy sample.

The angular power spectrum between redshift bin i of observable A and redshift bin j of observables B at projected Fourier mode l, $C^{ij}_{AB}(l)$ is computed using the Limber and flat sky approximations:

$$C^{ij}_{AB}(l) = \int d\chi \frac{q^i_A(\chi) q^j_B(\chi)}{\chi^2} P_{AB}(k = l/\chi, z(\chi)), \tag{1}$$

where χ is the comoving distance, $q^i_A(\chi)$ are weight functions of the different observables given in Eqs. (2-32), and $P_{AB}(k,z)$ the three dimensional, probe-specific power spectra detailed below. The weight function for the projected galaxy density in redshift bin i,$q^i_{\delta_g}(\chi)$, is given the normalized comoving distance probability of galaxies in this redshift bin

$$q^i_{\delta_g}(\chi) = \frac{n^i_{\rm lens}(z(\chi))}{\bar{n}^i_{\rm lens}} \frac{dz}{d\chi}, \tag{2}$$

with $n^i_{\rm lens}(z)$ the redshift distribution of galaxies in (photometric) galaxy redshift bin i (cf. Eq. (13)), and $\bar{n}^i_{\rm lens}$ the angular number densities of galaxies in this redshift bin (cf. Eq. (15)).

For the convergence field, the weight function $q^i_\kappa(\chi)$ is the lens efficiency,

$$q^i_\kappa(\chi) = \frac{3H_0^2\Omega_m}{2c^2} \frac{\chi}{a(\chi)} \int_\chi^{\chi_h} d\chi' \frac{n^i_{\rm source}(z(\chi'))dz/d\chi'}{\bar{n}^i_{\rm source}} \frac{\chi' - \chi}{\chi'}. \tag{3}$$

the lens efficiency, with $n^i_{\rm source}(z)$ the the redshift distribution of source galaxies in (photometric) source redshift bin i (Eq. (13)), $\bar{n}^i_{\rm source}$ the angular number densities of source galaxies in this redshift bin (Eq. (15)), and $a(\chi)$ the scale factor.

We model the probe specific three-dimensional power spectra $P_{AB}(k,z)$ as a function of the nonlinear matter density power spectrum $P_{\delta\delta}(k,z)$, for which we use the Takahashi et al. (2012) fitting formula

$$P_{\kappa\kappa}(k,z) = P_{\delta\delta}(k,z), \tag{4}$$

$$P_{\delta_g\kappa}(k,z) = b_g(z) P_{\delta\delta}(k,z), \tag{5}$$

$$P_{\delta_g\delta_g}(k,z) = b^2_g(z) P_{\delta\delta}(k,z). \tag{6}$$

We only consider the large-scale galaxy distribution and assume that the galaxy density contrast on these scales can be approximated as the nonlinear matter density contrast times an effective galaxy bias parameter $b_g(z)$. We defer the exact implementation details of the systematics to Sect. 4, where we discuss the specific analysis choices in the context of WFIRST and LSST.

2.3 Covariance and Inference Details

The multi-probe data vector, denoted as $\boldsymbol{D}$, is computed at the fiducial parameters in cosmology and systematics see Table 2. The same parameters enter in the computation of the non-Gaussian covariance matrix $\mathbf{C}$. Since this covariance matrix is calculated analytically and not estimated from either simulations or data, it does not inherently limit the number of data points that can enter our analysis (see e.g., Taylor et al. 2013; Dodelson & Schneider 2013, for details on these constraints).

We sample the joint parameter space of cosmological $\boldsymbol{p}_{\rm co}$ and nuisance parameters $\boldsymbol{p}_{\rm nu}$ and parameterize the joint likelihood as a multivariate Gaussian

$$L(\boldsymbol{D}|\boldsymbol{p}_{\rm co},\boldsymbol{p}_{\rm nu}) = N \times \exp\Big(-\frac{1}{2}\underbrace{\left[(\boldsymbol{D}-\boldsymbol{M})^t\,\mathbf{C}^{-1}\,(\boldsymbol{D}-\boldsymbol{M})\right]}_{\chi^2(\boldsymbol{p}_{\rm co},\boldsymbol{p}_{\rm nu})}\Big). \quad (7)$$

The model vector $\boldsymbol{M}$ is a function of cosmology and nuisance parameters, i.e. $\boldsymbol{M} = \boldsymbol{M}(\boldsymbol{p}_{\rm co},\boldsymbol{p}_{\rm nu})$ and the normalization constant $N = (2\pi)^{-\frac{n}{2}}|C|^{-\frac{1}{2}}$ can be ignored under the assumption that the covariance is constant in parameter space.

We calculate the covariance of two angular power spectra as the sum of the Gaussian covariance, $\mathrm{Cov}^{\rm G}$, and non-Gaussian covariance in the absence of survey window effects $\mathrm{Cov}^{\rm NG}$, and the super-sample covariance, $\mathrm{Cov}^{\rm SSC}$, which describes the uncertainty induced by large-scale density modes outside the survey window

$$\begin{aligned}\mathrm{Cov}\left(C^{ij}_{AB}(l_1),C^{ij}_{AB}(l_2)\right) &= \mathrm{Cov}^{\rm G}\left(C^{ij}_{AB}(l_1),C^{ij}_{AB}(l_2)\right)\\ &+ \mathrm{Cov}^{\rm NG}\left(C^{ij}_{AB}(l_1),C^{ij}_{AB}(l_2)\right)\\ &+ \mathrm{Cov}^{\rm SSC}\left(C^{ij}_{AB}(l_1),C^{ij}_{AB}(l_2)\right). \quad (8)\end{aligned}$$

We refer the interested reader to the Appendix (Eqs: A2-A14) of Krause & Eifler (2017) for the exact implementation of the 3 covariance terms.

The assumption of a constant, known covariance matrix $\mathbf{C}$ is an approximation to the correct approach of a cosmology dependent or estimated covariance (see Eifler et al. 2009; Sellentin & Heavens 2016, for further details).

Given the likelihood function we can compute the posterior probability in parameter space from Bayes' theorem and we sample the parameter space using `emcee`[10] (Foreman-Mackey et al. 2013), which is based on the affine-invariant sampler of Goodman & Weare (2010), and which can be parallelized with either MPI or shared memory multiprocessing.

3 SIMULATING WFIRST AND LSST SURVEYS

[All: Read, comment, suggest refs.
Rachel: read+edit LSST subsection.
Elisabeth, Xiao, Vivian: check eqs vs implementation.
Chris, Melanie, Shooby: read+edit WFIRST subsections]

[10] http://dan.iel.fm/emcee/current/

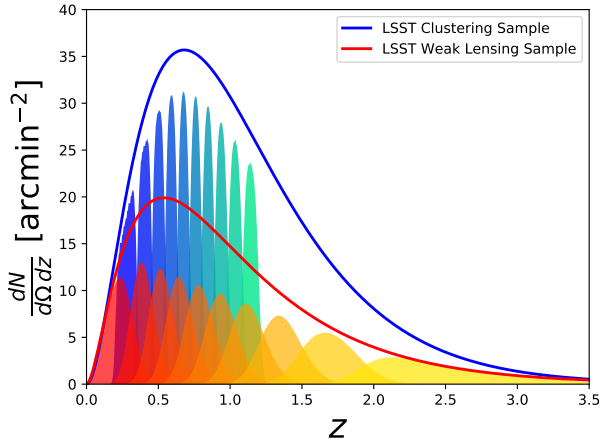


Figure 1. LSST redshift distribution with the 10 tomographic lens bins (blue) and 10 tomographic source bins (red), based on the ingredients from the DESC-SRD. We note the 10 lens bins being at lower redshift ($\leqslant 1.2$) and being more narrow compared to the WFIRST bins, which leads to a larger number of galaxy-galaxy lensing bins in the data vector.

3.1 LSST

In order to incorporate the latest analysis choices for a 3x2pt LSST analysis we closely follow The LSST Dark Energy Science Collaboration et al. (2018) (DESC-SRD18 from hereon) where the survey depths is based on the `OpSim` software[11]. The scenario described in the DESC-SRD18 has median depth (5σ point-source detection across the survey) of 25.30, 26.84, 27.04, 26.35, 25.22, 24.47 in ugrizy after 10 years.

These depths were determined after discarding areas with i-mag<26 to homogenize the depth across the whole survey after removing areas of high extinction near the Galactic plane. LSST dark energy science will exclude data from regions near a Galactic latitude of zero and likely discard areas with E(B-V)>0.2 . In DESC-SRD18 the resulting area after this homogenization process were 14,300 deg^2. For the purpose of this paper we however assume the more ambitious survey area of 18,000 deg^2, which is more demanding in terms of WFIRST observing time, if WFIRST were to cover the full LSST footprint.

We note that LSST is evaluating a variety of different survey strategies (see e.g., Lochner et al. 2018) with potential survey areas ranging from below 10,000 deg^2 to more than 25,000 deg^2 and that our 18,000 deg^2 extragalactic sky coverage scenario is well within the range of options.

Adopting these depth cuts and the associated galaxy selection function for source and lens samples we model the true redshift distribution $n_{\rm x}(z)$ for $x \in \{\text{lens, source}\}$ as

$$n_{\rm x}(z) \equiv \frac{d^2N_{\rm x}}{dz d\Omega_s} = \bar{n}_{\rm x}\frac{\Theta(z_{\rm max}-z)\,z^2\exp\left[-\left(\frac{z}{z_0}\right)^\alpha\right]}{\int_0^{z_{\rm max}} dz\,z^2\exp\left[-\left(\frac{z}{z_0}\right)^\alpha\right]}, \quad (9)$$

$N_{\rm x}$ denotes the total number of source/lens galaxies, and $\bar{n}_{\rm x}$ the effective number density of source/lens galaxies. Two relevant parameters determine the functional form of the distribution, i.e. $(z_0 = 0.28, \alpha = 0.9)$. We impose a high-$z$ cut $z_{\rm max} = 3.5$ for sources and $z_{\rm max} = 1.2$ for lenses, following the analysis choices of DESC-SRD18 and we yield the following number densities for lenses and

[11] https://github.com/lsst/sims_operations

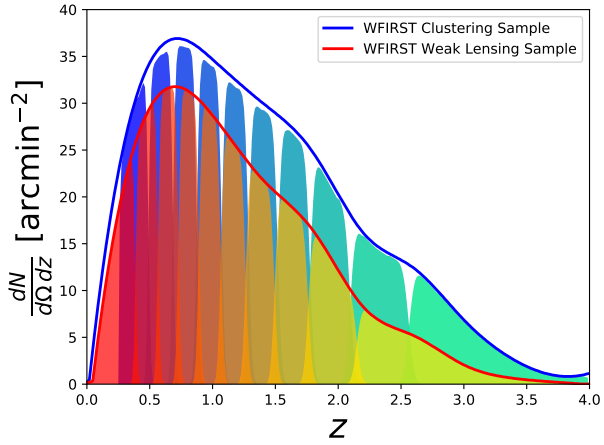


Figure 2. This plot shows the WFIRST redshift distribution with the 10 tomographic lens bins (blue) and 10 tomographic source bins (red). The Gaussian photo-z tomographic bin-width ($\sigma_z = 0.01$) corresponds to the optimistic scenario that we assume for the combined WFIRST HLS and LSST band coverage, i.e. 4 NIR and 6 optical bands.

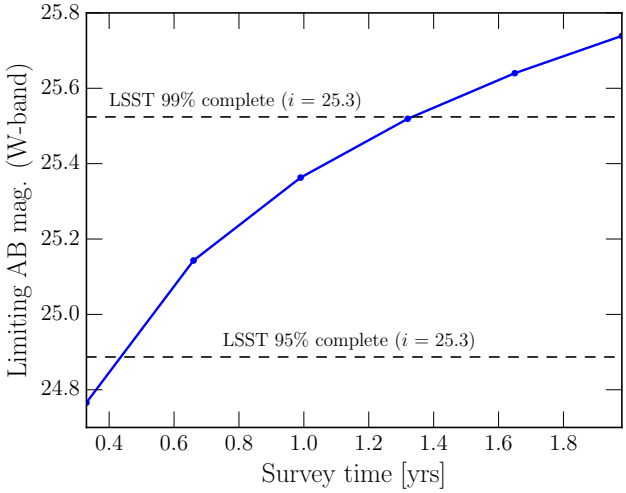


Figure 3. Limiting magnitude of a 18,000 deg^2 WFIRST W-band survey as a function of survey time. We also show the LSST weak lensing samples 95% and 99% completeness thresholds as dashed lines.

sources

$$\bar{n}_{\rm source} = N_{\rm source}/\Omega_{\rm s} = 27\ {\rm galaxies/arcmin}^2 \quad (10)$$

$$\bar{n}_{\rm lens} = N_{\rm lens}/\Omega_{\rm s} = 48\ {\rm galaxies/arcmin}^2\,. \quad (11)$$

The lens galaxies correspond to the LSST gold sample, the source galaxy number density was obtained after removal of masked and seriously blended objects. For both lenses and sources we define 10 tomographic bins. For the sources the bin limits are chosen such that $\bar{n}^i_{\rm source} = 2.7$. For the lens sample, we choose equidistant bins of 0.1 between 0.2-1.2, which implies that the number density per bin is defined as

$$\bar{n}^i_x = \int dz\, n^i_x(z). \quad (12)$$

The true redshift distribution in Eq. (9) is then convolved with a Gaussian photometric redshift uncertainty model to obtain the distribution within tomographic bin i

$$n^i_x(z_{\rm ph}) = \int_{z^i_{{\rm min},x}}^{z^i_{{\rm max},x}} dz\, n_{\rm x}(z)\, p^i\left(z_{\rm ph}|z,x\right), \quad (13)$$

where $p\left(z_{\rm ph}|z,x\right)$ is the probability distribution of $z_{\rm ph}$ at given true redshift z for galaxies from population x

$$p^i\left(z_{\rm ph}|z,x\right) = \frac{1}{\sqrt{2\pi}\sigma_{z,x}(1+z)} \exp\left[-\frac{\left(z-z_{\rm ph}-\Delta^i_{z,x}\right)^2}{2\left(\sigma_{z,x}(1+z)\right)^2}\right]. \quad (14)$$

The resulting Gaussian tomographic bin is parametrized through scatter $\sigma_z(z)$ and bias between $z - z_{\rm ph}$, i.e. $\Delta^i_z(z)$. The bias $\Delta^i_z(z)$ has fiducial value of zero, for the lens sample $\sigma_z = 0.03$ for the sources $\sigma_z = 0.05$. The resulting distributions are shown in Fig. 1. We define uncertainties in the redshift distribution in Sect. 4.2.

3.2 WFIRST

We use the WFIRST exposure time calculator (ETC) in weak lensing mode (Hirata et al. 2012) to compute realistic survey scenarios for WFIRST's coverage of area and depth in a given band. We fix the time per exposure and vary the number of exposures to build up depth over the survey area of a given scenario. For the HLS survey this area is 2000 deg^2, for the WFIRST wide scenario it corresponds to the LSST area of 18,000 deg^2. The total survey time for a given number of exposures includes a simple prescription for overheads and is correct to approximately 10%.

In order to obtain accurate redshift distributions we closely follow Hemmati et al. (2018) in applying the ETC results to the CANDELS data set (Grogin et al. 2011), which is the only data set available that is sufficiently deep to model WFIRST observations. The ETC has a built-in option to obtain a weak lensing catalog based on an input catalog of detected sources. The criteria for galaxies to be considered weak lensing suitable are S/N > 18 (J+H band combined, matched filter), ellipticity dispersion $\sigma_\epsilon < 0.2$, and resolution factor $R > 0.4$, where and resolution factor are in Bernstein & Jarvis (2002) convention (i.e. "$\epsilon = (a^2 - b^2)/(a^2 + b^2)$ instead of $(a-b)/(a+b)$).

We apply this feature to the CANDELS catalog and generate the lensing sample for the WFIRST HLS 4 NIR band survey. For the lens sample we select CANDELS galaxies with S/N>10 in the 4 WFIRST bands. Our WFIRST analysis assumes LSST photometry from the ground, hence we further down-select both samples by imposing a S/N>5 cut in each LSST band.

The resulting number densities for the HLS are

$$\bar{n}_{\rm source} = N_{\rm source}/\Omega_{\rm s} = 51\ {\rm galaxies/arcmin}^2 \quad (15)$$

$$\bar{n}_{\rm lens} = N_{\rm lens}/\Omega_{\rm s} = 66\ {\rm galaxies/arcmin}^2\,. \quad (16)$$

Based on these lens and source samples we define 10 tomographic bins such that $\bar{n}^i_x = \bar{n}_x/10$. These tomographic bins are then convolved with a Gaussian distribution (see Eq. (13)) with mean zero and $\sigma_z = 0.01$. This optimistic, narrow width of the Gaussian kernel is motivated by the fact that our HLS survey assumes a joint WFIRST and LSST sample, which altogether gives good photometry in 10 bands ranging from 0.3-2 micons. The resulting redshift distributions are depicted in Fig. 2.

3.3 WFIRST survey strategy variations

After defining the LSST and the WFRIST HLS scenarios in the past sections we now explore variations in the WFIRST survey strategy, in particular a WFIRST wide scenario that covers the LSST footprint in the W-band (see Table 1). We again use the CANDELS

catalog when defining galaxy samples for WFIRST and LSST below.

Figure 3 shows the results when using the ETC to compute the depth of the W-band and of the H-band as a function of time under the assumption that no other bands are used. We find that a ~5 month WFIRST W-band survey can obtain high-resolution space imaging for ~95% of the LSST weak lensing sample (see Fig. 3. We define "LSST weak lensing" sample as galaxies that have a 20σ point detection in the LSST i-band.

As a first result of this paper we conclude that if blending poses a systematics limitation to LSST weak lensing cosmology, a dedicated 5 month WFIRST survey would identify almost all LSST blends and enable modeling of shapes and photo-z for said blends.

A 1.3 year WFIRST W-band survey will provide corresponding information for ~99% of the LSST weak lensing sample and of course also substantially increase the depth of the WFIRST imaging, which opens up the idea to use the deeper WFIRST imaging for shape measurements and combine these with the ground based LSST photometry.

We also define a WFIRST wide scenario for the full simulated likelihood that we will use to compare to the LSST and to the WFIRST HLS surveys. We assume a 1.5 year WFIRST wide survey in the W-band and follow same procedure as for the WFIRST HLS survey (Sect. 3.2) in deriving the lens and source sample.

Figure 4 shows the number density of galaxies suitable for shape measurements from a WFIRST 18,000 deg^2 as a function of survey time and Fig. 5 shows the corresponding fraction of LSST galaxies for which good photo-z information (5σ detection in all LSST bands) can be obtained. We also show the corresponding results for the WFIRST H-band, which is useful as an alternative to the W-band since wavelength dependent Point-Spread Function (PSF) modeling might prohibit shape measurements from a band as broad as the W-band. WFIRST's diffraction-limited PSF size ranges from 0.085" to 0.175" over the W-band, which is about a 50% change, compared to only a 20% change when using WFIRST's H-band.

We note that the ESA/NASA Euclid satellite mission is developing mitigation techniques for a similar problem given that Euclid's diffraction-limited PSF size ranges from 0.085" to 0.155" over the VIS band, i.e. the main band in which Euclid measures shapes. Cypriano et al. (2010); Carlsten et al. (2018); Eriksen & Hoekstra (2018) propose a variety of methods how to control wavelength dependent PSF uncertainties through a combination of improved galaxy spectral energy templates or PSF measurements based on stars that span the same color range as the galaxies. Additional photometric information, e.g. from the ground is the main avenue for Euclid to gain the relevant information to mitigate this effect, albeit this is of course limited by the difference in resolution of space and ground imaging (also see Meyers & Burchat 2015, for additional wavelength dependent PSF effects from the atmosphere). WFIRST is in the unique position to collect narrow band, space resolution imaging over a smaller, but representative area and calibrate it's wide W-band survey if this effect becomes the dominant systematic.

A 1.5 year WFIRST wide survey would yield 45 galaxies/arcmin^2 for the source sample (cf Fig. 4) and 68 galaxies/arcmin^2 for the lens sample, which is again defined as a S/N>10 cut based on the CANDELS catalog. Since we require good LSST photometry for our WFIRST galaxy sample these number densities are further reduced to 43 galaxies/arcmin^2 (cf Fig. 5) for the joint source and 50 galaxies/arcmin^2 for the joint lens sample.

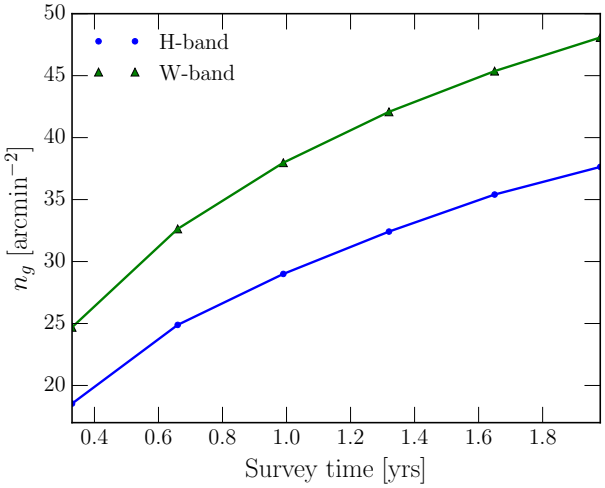


Figure 4. The number density of a weak lensing galaxy sample for a 18,000 deg^2 WFIRST survey when conducted in W or H-band, respectively, again as a function of survey time.

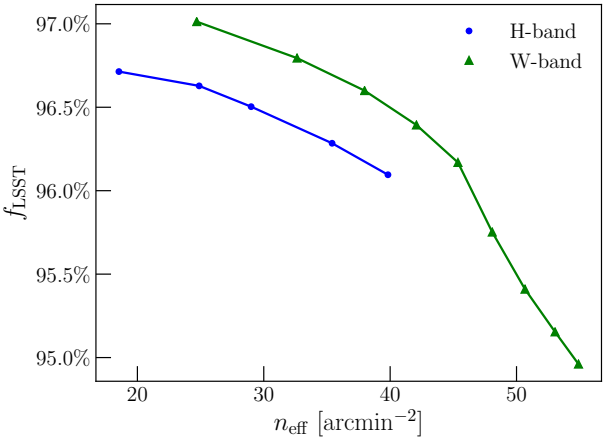


Figure 5. Fraction of LSST galaxies with acceptable multi-band photometry as a function of number density of a WFIRST weak lensing sample, based on the CANDELS catalog.

The calculation of the redshift distributions follow the same procedure as for the WFIRST HLS survey (Sect. 3.2), the only difference being that we assume a slightly wider Gaussian kernel $\sigma_z = 0.02$ compared to the HLS scenario.

4 LIKELIHOOD ANALYSIS - WEAK LENSING AND GALAXY CLUSTERING

[All: Read, comment, suggest refs.
Hung-Jin: read+edit baryons
Elisabeth : check IA eqs vs implementation
Xiao/Jiachuan/Paul: check chain convergence section, FoM difference computation]

In Sects. 2 and 3 we describe the basic setup of our analysis including covariance computation, modeling of observables, inference process, galaxy sample selection, and redshift distribution computation. In the following we detail the analysis choices for our likelihood analysis including details on the systematics implementation.

Table 2. Fiducial parameters, flat priors (min, max), and Gaussian priors (μ, σ).

	WFIRST HLS		**LSST**		**WFIRST wide**	
Parameter	Fid	Prior	Fid	Prior	Fid	Prior
			Survey			
Ω_s	2,000 deg^2	fixed	18,000 deg^2	fixed	18,000 deg^2	fixed
$n_{\rm source}$	51 gal/arcmin^2	fixed	27 gal/arcmin^2	fixed	43 gal/arcmin^2	fixed
σ_ϵ		0.26			fixed	
$n_{\rm lens}$	66 gal/arcmin^2	fixed	48 gal/arcmin^2	fixed	50 gal/arcmin^2	fixed
			Lens photo-z			
$\Delta^i_{\rm z,lens}$	0.0	Gauss (0.0, 0.001)	0.0	Gauss (0.0, 0.001)	0.0	Gauss (0.0, 0.001)
$\sigma_{\rm z,lens}$	0.01	Gauss (0.01, 0.002)	0.03	Gauss (0.03, 0.003)	0.02	Gauss (0.02, 0.002)
			Source photo-z			
$\Delta^i_{\rm z,source}$	0.0	Gauss (0.0, 0.001)	0.0	Gauss (0.0, 0.001)	0.0	Gauss (0.0, 0.001)
$\sigma_{\rm z,source}$	0.01	Gauss (0.01, 0.002)	0.05	Gauss (0.05, 0.003)	0.05	Gauss (0.02, 0.002)
			Shear calibration			
m_i	0.0	Gauss (0.0, 0.002)	0.0	Gauss (0.0, 0.003)	0.0	Gauss (0.0, 0.002)

Common assumptions across all surveys		
Parameter	Fid	Prior
	Cosmology	
$\Omega_{\rm m}$	0.3156	flat (0.1, 0.6)
σ_8	0.831	flat (0.6, 0.95)
n_s	0.9645	flat (0.85, 1.06)
w_0	-1.0	flat (-2.0, 0.0)
w_a	0.0	flat (-2.5, 2.5)
$\Omega_{\rm b}$	0.0492	flat (0.04, 0.055)
h_0	0.6727	flat (0.6, 0.76)
	Galaxy bias (tomographic bins)	
$b^i_{\rm g}$	1.3 + i $\times$ 0.1	flat (0.8, 3.0)
	Intrinsic alignment	
$A_{\rm IA}$	5.95	flat (0.0, 10)
$\beta_{\rm IA}$	1.1	flat (-4.0, 6.0)
$\eta_{\rm IA}$	0.49	flat (-10.0, 10.0)
$\eta^{\rm high-z}_{\rm IA}$	0.0	flat (-1.0, 1.0)
	Baryonic Physics	
Q_1	0.0	Gauss (0.0, 16.0)
Q_2	0.0	Gauss (0.0, 5.0)
Q_3	0.0	Gauss (0.0, 0.8)

4.1 Building a 3x2pt data vector

Source galaxies – cosmic shear Given the 10 tomographic bins for the source sample we compute 55 cosmic shear auto-and cross power spectra, which we divide into 15 logarithmically spaced Fourier mode bins ranging from $l_{\rm min} = 30$ to $l_{\rm max} = 3000$.

Lens galaxies – galaxy clustering The galaxy clustering data vector is also divided into 15 l-bins ranging from 30-3000, however we exclude high $l-$bins, if scales below $R_{\rm min} = k_{\rm max}/2\pi = 21$Mpc/h contribute to the projected integral (see Eq. (1)).

Lens $\times$ source galaxies – galaxy-galaxy lensing The galaxy-galaxy lensing part of the data vector assumes the lens galaxy sample as foreground and the source galaxy sample as background galaxies; we only consider non-overlapping source and lens in redshift bins. We again impose a cut-off at $R_{\rm min} = 21$Mpc/h for the baseline model. We note that the low, narrow redshift distribution of the 10 lens tomographic bins of the LSST scenario leads to a substantially higher number of galaxy-galaxy lensing power spectra (52), compared to both WFIRST scenarios (32).

4.2 Systematics

We parameterize uncertainties arising from systematics through nuisance parameters, which are summarized with their fiducial values and priors in Table 2. Our default likelihood analysis includes the following systematics:

Photometric redshift uncertainties As described in detail in Sect. 3 we consider Gaussian photometric redshift uncertainties, which are characterized by scatter $\sigma_z(z)$ and bias $\Delta_z(z)$. While these may in general be arbitrary functions, we further assume that the scatter can be described by the simple redshift scaling $\sigma_{z,x}(1+z)$ and allow one (constant) bias parameter $\Delta^i_{z,x}$ per redshift bin (cf Eq. (14)).

For all our likelihood analyses we have 10 tomographic bins for the lens and source sample, hence 20 photo-z bias parameters $\Delta^i_{z,x}$ and two photo-z scatter parameters $\sigma_{z,x}(1+z)$. We note that the selection and characterization of lens and source galaxy samples are some of the most challenging choices in a multi-probe analysis and a variable that warrants future investigation. The trade space of photo-z accuracy vs number density, e.g. by using a redmagic sample (Rozo et al. 2016) instead of the LSST Gold sample is interesting to explore further.

Linear galaxy bias is described by one nuisance parameter per lens galaxy redshift bin b^i_g, which is marginalized over using conservative flat priors $[0.8; 3.0]$.

We note that the fiducial values of the galaxy bias parameters have little impact on the results, however that a more stringent prior would be highly beneficial. For example, the interplay of galaxy bias and photo-z uncertainties limits the ability of galaxy-galaxy lensing to self-calibrate intrinsic alignment models. Prior information on linear galaxy bias parameters would also help implement high-order bias models that require additionial free parameters.

Shape measurement uncertainties are modeled as multiplicative shear calibration. We use one parameter m^i per redshift bin, which affects cosmic shear and galaxy-galaxy lensing power spectra as

$$\begin{aligned} C^{ij}_{\kappa\kappa}(l) &\longrightarrow (1+m^i)(1+m^j)C^{ij}_{\kappa\kappa}(l), \\ C^{ij}_{\delta_g\kappa}(l) &\longrightarrow (1+m^j)C^{ij}_{\delta_g\kappa}(l), \end{aligned} \tag{17}$$

The fiducial value of each m^i is zero and we marginalize over the m^i independently with Gaussian priors of different width for LSST and WFIRST. Altogether shape uncertainties add 10 nuisance parameters to our analysis.

Intrinsic alignment Intrinsic alignment (IA) of source galaxies has been studied extensively as a systematic for weak lensing through observations, simulations, and theory (e.g., Hirata & Seljak 2004; Mandelbaum et al. 2006; Joachimi & Bridle 2010; Singh et al. 2015; Troxel & Ishak 2014; Tenneti et al. 2015; Blazek et al. 2015)

The assumption that the shape and orientation of an elliptical galaxy are determined by the shape of the halo in which it resides causes a correlation of intrinsic galaxy ellipticities with the gravitational tidal field. These "tidal alignment" models are frequently used to describe IA of elliptical (red) galaxies and cause the dominant IA effects. We do not consider quadratic alignment models or more sophisticated IA modeling as a function of galaxy color.

Our implementation follows Krause et al. (2016) for cosmic shear and Krause & Eifler (2017) for the extension to galaxy-galaxy lensing. The cosmic shear and galaxy-galaxy lensing projected power spectra are modified by additive terms to include the IA effects, specifically

$$C^{ij}_{\kappa\kappa}(l) \rightarrow C^{ij}_{\kappa\kappa}(l) + C^{ij}_{\rm II}(l) + C^{ij}_{\rm GI}(l), \tag{18}$$

$$C^{ij}_{\delta_g\kappa}(l) \rightarrow C^{ij}_{\delta_g\kappa}(l) + C^{ij}_{\delta_g{\rm I}}(l). \tag{19}$$

The $C^{ij}_{\rm II}(l)$ term arises since galaxies that are physically close experience an alignment of their intrinsic ellipticity (II) from the same tidal field. For galaxy pairs that are separated in redshift alignment effects also occur: foreground galaxies are aligned by the same tidal field that lenses background galaxies, which introduces a (anti-) correlation of shear (G) and intrinsic (I) ellipticity.

The $C^{ij}_{\delta_g{\rm I}}(l)$ effect describes a correlation of galaxy overdensity and intrinsic ellipticity (I) for galaxy pairs that are physically close and affected by the same tidal field.

The projected power spectra can be computed from the corresponding 3D power spectra as described in Eq. (1), specifically

$$C^{ij}_{\rm II}(l) = \int {\rm d}\chi \frac{q^i_{\delta_g}(\chi) q^j_{\delta_g}(\chi)}{\chi^2} f^2_{\rm red}(m_{\rm lim}, z) P_{\rm II}(k,z), \tag{20}$$

$$C^{ij}_{\rm GI}(l) = \int {\rm d}\chi \frac{q^i_{\delta_g}(\chi) q^j_{\kappa}(\chi)}{\chi^2} f_{\rm red}(m_{\rm lim}, z) P_{\rm GI}(k,z), \tag{21}$$

$$C^{ij}_{\delta_g{\rm I}}(l) = \int {\rm d}\chi \frac{q^i_{\delta_g}(\chi) q^j_{\kappa}(\chi)}{\chi^2} f_{\rm red}(m_{\rm lim}, z) P_{\delta_g{\rm I}}(k,z), \tag{22}$$

where $f_{\rm red}$, the fraction of red galaxies at redshift $z(\chi)$, which is evaluated from the GAMA luminosity function assuming a limiting magnitude $m_{\rm lim} = 25.3$, which corresponds to our definition of the LSST weak lensing sample (cf. Fig. 3).

In the tidal alignment picture, where to leading order the intrinsic ellipticity field is linear in the density field we can write the relevant 3D-IA power spectra as

$$P_{\rm II}(k,z) = A^2(m_{\rm lim}, z) P_{\delta\delta}(k,z), \tag{23}$$

$$P_{\rm GI}(k,z) = -A(m_{\rm lim}, z) P_{\delta\delta}(k,z), \tag{24}$$

$$P_{\delta_g{\rm I}}(k,z) = -A(m_{\rm lim}, z) b_g(z) P_{\delta\delta}(k,z). \tag{25}$$

with $m_{\rm lim} = 25.3$ as the survey's limiting magnitude. Uncertainties in galaxy bias in Eq. (25) are again parameterized as one free parameter per tomographic lens bin.

Our likelihood analysis includes uncertainties in IA modeling via four parameters that enter the IA amplitude $A(L,z)$ (see Table 2 for details).

We compute the average IA amplitude from the red galaxy luminosity distribution from the GAMA survey and include an additional parameter $\eta_{\rm high-z}$ to account for the uncertainty in the extrapolation of this redshift scaling,

$$A(m_{\rm lim}, z) = \left\langle A(L,z) \right\rangle_{\phi_{\rm red}} \times \left[\Theta(z_1 - z) + \Theta(z - z_1) \left(\frac{1+z}{1+z_1} \right)^{\eta_{\rm high-z}} \right], \tag{26}$$

where Θ is the step function, which separates the range where our fiducial redshift scaling is based on the MegaZ-LRG + SDSS LRG sample Joachimi et al. (2011) and where we extrapolate this functional form. The term in square brackets is a truncated power-law in $(1+z)$ with an additional uncertainty parameterized as $\eta_{\rm high-z}$ for $z > z_1$. Given that the MegaZ-LRG + SDSS LRG extends to $z \leqslant 0.7$ we choose $z_1 = 0.7$.

The other three nuisance parameters A_0, β, η enter via

$$A(L,z) = \frac{C_1 \rho_{\rm m,0}}{D(z)} A_0 \left(\frac{L}{L_0} \right)^{\beta} \left(\frac{1+z}{1+z_0} \right)^{\eta}. \tag{27}$$

In this expression $C_1 \rho_{\rm cr} = 0.0134$ is derived from SuperCOSMOS observations (Hirata & Seljak 2004; Bridle & King 2007), and for the fit parameters $A, \eta_{\rm other}, \beta$ we adopt the constraints from the MegaZ-LRG + SDSS LRG sample in, i.e. $A_0 = 5.92^{+0.77}_{-0.75}$,

$\eta = -0.47^{+0.93}_{-0.96}$, and $\beta = 1.10^{+0.29}_{-0.30}$, with $z_0 = 0.3$ the observationally-motivated pivot redshift and L_0 the pivot luminosity corresponding to an absolute r-band magnitude of -22.

We do not consider additional uncertainties in the luminosity function (cf Krause et al. 2016, where said uncertainties are marginalized over 6 additional parameters), but we note that these uncertainties can be significant. This is the interface where cosmology meets galaxy formation and we acknowledge that it is critical to understand the latter to precisely constrain the former.

Baryonic physics Baryonic physics affecting the modeling of small scales in the matter power spectrum (e.g. van Daalen et al. 2011; Chisari et al. 2018) are a pressing concern for cosmic shear (see e.g. Semboloni et al. 2011; Zentner et al. 2013; Eifler et al. 2015; Huang et al. 2019) and will become a pressing concern for galaxy-galaxy lensing and galaxy clustering if higher-order bias models are included that allow pushing to smaller scales.

We mitigate the impact of baryonic physics in two different ways (cf. Chisari et al. 2019, for an overview on mitigation strategies): First, the scale cuts for the galaxy clustering and galaxy-galaxy lensing part of the data vector that are driven by our assumption of using linear galaxy bias, i.e. $R_{\rm min} = 21$Mpc/h are conservative and probably sufficient to control baryonic effects for these parts of the data vector. For cosmic shear, our $l_{\rm max} = 3000$ cut also mitigates the impact of baryonic physics but as (Huang et al. 2019, (H19 from hereon)) have shown this is insufficient as a scale cut to control baryons at the LSST Year 10 level.

We employ "Method C" detailed in H19 to account for residual uncertainties in baryonic physics in our full 3x2pt data vector. In short, we compute the difference of dark matter only to baryonic 3x2 data vectors for LSST and WFIRST for 5 different baryonic scenarios. The baryonic scenarios are extracted from hydro-simulations, in particular we use the MassiveBlack-II simulation (Khandai et al. 2015), IllustrisTNG (Weinberger et al. 2018; Pillepich et al. 2018), Horizon-AGN (Dubois et al. 2016), Eagle simulation (Schaye et al. 2015) , and the OWLS AGN simulation (Schaye et al. 2010; van Daalen et al. 2011). These choice of simulations is motivated by the fact that we require AGN feedback to be included and that we require sufficiently high resolution to reliably mode small scales (again see H19 for a summary of the simulations and motivation of this choice).

From the 5 baryonic scenarios we extract 3x2pt data vectors with the exact properties as discussed in Sect.4.1 and we build a "difference matrix" of baryonic and dark matter data vectors (see Eq. (20) in H19)

$$\boldsymbol{\Delta}(\boldsymbol{p}_{\rm co}) = \left[\begin{array}{ccc} | & & | \\ \boldsymbol{B}_1 - \boldsymbol{M} & \dots & \boldsymbol{B}_5 - \boldsymbol{M} \\ | & & | \end{array} \right]_{N_{\rm data} \times N_{\rm sim}} \tag{28}$$

We weigh this difference matrix with respect to the statistical uncertainty described in our covariance matrix $\mathbf{C}$, i.e. we perform a cholesky decomposition $\mathbf{C} = \mathbf{L}^{-1}\mathbf{L}$ and compute

$$\boldsymbol{\Delta}_{\rm ch}(\boldsymbol{p}_{\rm co}) = \mathbf{L}^{-1}\boldsymbol{\Delta}(\boldsymbol{p}_{\rm co}) = \mathbf{U}_{\rm ch}\, \boldsymbol{\Sigma}_{\rm ch}\, \mathbf{V}^{\rm t}_{\rm ch}\,, \tag{29}$$

where in the last step we perform an singular value decomposition on the weighted difference matrix to extract the Principal Components (PCs) that span the range of uncertainty in baryonic physics. This PC basis set is stored in $\mathbf{U}_{\rm ch}(\boldsymbol{p}_{\rm co})$

$$\boldsymbol{\Delta}_{\rm ch}(\boldsymbol{p}_{\rm co}) = \mathbf{L}^{-1}(\boldsymbol{B} - \boldsymbol{M}) = \sum_{n=1}^{5} Q_n\, \mathbf{PC}_n\,. \tag{30}$$

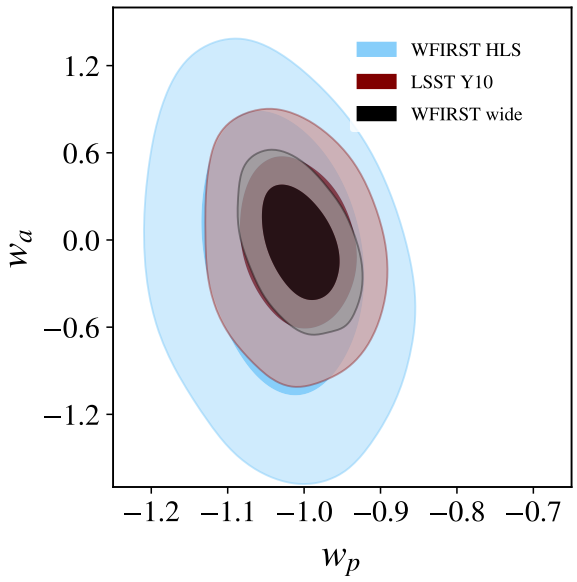


Figure 6. Gain in constraining power on dark energy parameters w_0 and w_a, marginalized over 5 other cosmological parameters and 49 nuisance parameters describing observational and astrophysical systematics. We show results for the 18,000 deg^2 LSST Year 10 data set in *red*, for the 2000 deg^2 WFIRST HLS survey in *blue*, and for the WFIRST wide survey *black*. Both WFIRST scenarios assume LSST multi-band photometry over the corresponding area.

By reorganizing Eq. (30), we can generate model vectors that include dark matter and baryonic physics as

$$\boldsymbol{M}(\boldsymbol{p}_{\rm co}, Q) = \boldsymbol{M}(\boldsymbol{p}_{\rm co}) + L \sum_{n=1}^{m} Q_n\, \mathbf{PC}_n\,, \tag{31}$$

where $m \leqq 5$. We include baryons in our analysis by replacing $\boldsymbol{M}$ in Eq. (7) with Eq. (31). We note that the cosmology dependence only enters through the dark matter part of the model vector, while the amplitudes of PCs are used as higher order correction for baryonic effects. We include the first 3 PCs in our likelihood analysis and consequently marginalize over 3 amplitudes Q as additional degrees of freedom to model baryonic physics.

The priors for the Q are highly conservative and chosen such that the 1σ region of the Gaussian prior corresponds to twice the amplitude of Q's needed to capture the Illustris (not TNG) simulation. The original Illustris simulation has a very strong feedback scenario which is already highly unlikely given present observations.

4.3 Simulated likelihood analysis

Using the data vectors defined Sect. 4.1 and the analysis choices defined in Table 2 we simulate likelihood analyses for LSST, WFIRST HLS, and WFIRST wide acknowledging that the latter two include the LSST data set. Altogether our likelihood analyses span 56 dimensions, 7 of which relate to cosmological parameters and the remaining 49 describe uncertainties in modeling observational (shear calibration, lens and source photo-z uncertainties) and astrophysical systematics (galaxy bias, intrinsic alignment, baryonic physics).

We use the fast CosmoLike forecasting modules and the `emcee` parallel sampling algorithm to generate chains with 1120 walkers,

8000 steps each, altogether our chains comprise 8.96M steps. To test its sampling convergence, we compare sub-chains of 140,000 steps in starting at step 2.1M and find that our FoMs have stabilized at 6M steps (cf Fig. **??**). We also vary the number of walkers and starting points (including their variance) and found our results independent of reasonable choices in these settings.

Figure 6 shows a substantial increase in the ability to constrain time-dependent dark energy for the WFIRST wide scenario compared to the LSST-only scenario and compared to the WFIRST HLS survey. We re-emphasize that both WFIRST scenarios assume existing LSST data over the corresponding area.

Regarding the WFIRST HLS vs WFIRST wide comparison (blue vs black contours in Fig. 6) the gain in constraining power (Δ FoM=???????) is mostly driven by the increased area 18,000 deg^2 vs 2,000 deg^2. The larger number density of lens and source galaxies and the improved photo-z accuracy cannot compensate for this increase. It is interesting to see that at the precision of WFIRST HLS and WFIRST wide the results are not systematics dominated but that an increase in area has a substantial impact on the contours.

Looking more closely at the composition of statistical uncertainties it is important to note that an increase in number density of galaxies reduces the noise terms in the covariances, especially the shape noise terms in the cosmic shear part of the covariance. It does not affect the signal part of the covariance, in particular the connected covariance terms and the Super-Sample Variance terms are only reduced with increasing area. The statement that area is more important than number density finds its limits if dark energy is a strong local function in redshift. In order to find such a localized redshift dependent dark energy feature a more narrow binning of the tomographic bins is required, which reduces the number of galaxies per tomographic bin, which increases the importance of the noise terms in the covariance. Depending on the number of tomographic bins number density of galaxies could be more important than area.

Regarding the LSST vs WFIRST wide comparison (red vs black contours in Fig. 6) the increase in constraining power (Δ FoM=???????) stems from a combination of increase in number density, deeper redshift distribution, improved control of systematics, most importantly photo-z calibration. A more detailed study is required to evaluate the exact trade space, in particular since photo-z parameters are highly correlated with our ability to model galaxy bias and intrinsic alignment. Such a study should also include more realistic photo-z errors, such as catastrophic outliers and it should investigate higher-order bias models to push to smaller scales in the galaxy-galaxy lensing and galaxy clustering parts of the data vector.

It is interesting to compare the LSST and WFIRST HLS contours (blue vs red contours in Fig. 6) given that these differ most significantly in terms of assumptions that enter the likelihood analysis. The substantial difference in area 18,000 deg^2 for LSST vs 2,000 deg^2 for WFIRST HLS is countered by a significantly higher number density of galaxies (???vs???), deeper redshift distribution, higher photo-z and shape measurement accuracy for the WFIRST HLS scenario. Despite these improvements the increase in area is more important and favors the LSST only scenario, albeit not at a level one might have suspected. It is also important to note that the WFIRST HLS data is collected in only 1.6 years of observing time, whereas the LSST data set assumes 10 years of observations. Unsurprisingly, systematics control is an important factor in future likelihood analyses and it is the exact reason why combining multiple surveys is a powerful concept.

5 DISCUSSION

[All: Read, comment, suggest refs.
Chris: check WFIRST implementation paragraph.
David: check "beyond weak lensing and galaxy clustering" subsect.
Cyrille, Bhuv, Chen, Vivian: Check CMB paragraph.
] In terms of how to fit the wide survey into the WFIRST mission, several options arise:

(i) WFIRST does not have expendables that would prohibit extended observations beyond its 5-year primary mission. Dedicating 1.5 years of such an extension to a W-band survey as detailed above, which would be well-matched in terms of timescale to LSST Y10, is a possible scenario.

(ii) A second option is to get the wide survey data earlier and to reduce the 2000 deg^2 footprint of the HLS in the primary mission. As an extreme example one could even fit the 1.5 year W-band survey outlined above into the nominal 1.6 year HLS survey. This however would come at the very high cost of having almost no grism or multi-band WFIRST coverage; given how important corresponding data is for systematics control this option appears less favorable.

(iii) A third option would be to cover a subset of the 18,000 deg^2 with the W-band in the WFIRST primary mission, e.g. one could survey 10,400 deg^2 with E(B-V)<0.08 that pass within 32 degrees of zenith at LSST. This will likely be the best part of the LSST footprint for extragalactic studies, and it can be surveyed in approximately 1 year to the same depth as our 1.5year fiducial survey.

We again refer to Fig. 3, as one of the main results of our paper, which shows that already a 5-month WFIRST survey in the W-band can provide 18,000 deg^2 high-resolution imaging for >95% of the LSST Year 10 weak lensing galaxy sample. For the 10,400 deg^2 with E(B-V)<0.08 this type of survey would only take 14 weeks. It is important to note that the WFIRST observing time, including the survey design of the HLS, has not been allocated to date, and that corresponding decisions will depend on the science landscape in ~5 years. In particular, it will be important to see how much systematics impact the first 3 years of LSST data and how much core science interests in the community (not just cosmology) would benefit from rapid WFIRST wide coverage.

This paper does not recommend replacing the current 2000 deg^2 HLS survey with a wide W-band survey of WFIRST. The current design of the HLS ensures exquisite systematics control and it is the consensus in the community that systematics control will be more important than maximizing statistical power. We instead recommend the exploration of the WFIRST wide survey strategies in combination with the HLS approach, specifically we envision that a wide WFIRST survey component would use the reference HLS to anchor shear and photo-z calibration.

5.1 WFIRST wide synergies beyond weak lensing and galaxy clustering

A WFIRST wide survey in one band as described above opens numerous science applications in cosmology beyond the joint weak lensing and galaxy clustering dark energy science depicted in Fig. 5.

Galaxy cluster science: An 18,000 deg^2 W-band survey would substantially enhance the weak lensing mass calibration of clus-

ters, a key ingredient for cluster cosmology (von der Linden et al. 2014a,b; Mantz et al. 2015; Miyatake et al. 2019; Costanzi et al. 2019). Both the increased spatial resolution and the broader wavelength coverage will help to decrease systematic uncertainties due to blending and photo-z misclassification, which are amplified in overdense cluster fields. An important aspect is that the W-band addition would enable precise mass calibration to higher redshift clusters than LSST alone (cf. Schrabback et al. 2018, for a corresponding endeavor using the 100 times smaller FoV of the HST).

Trough cosmology: Cosmic underdensities have emerged as an interesting probe of structure formation and hence dark energy and modified gravity (Krause et al. 2013; Melchior et al. 2014; Hamaus et al. 2016; Davies et al. 2019); potentially corresponding observables are easier to model compared to probes that rely on higher-density environments. Trough cosmology (Gruen et al. 2016, 2018), and especially trough lensing, would benefit from the 18,000 deg^2 wide WFIRST survey scenario because of the higher density of source galaxies (for trough lensing) and the higher density of detected galaxies that signify the trough boundaries.

Strong Lensing: LSST will find a large number of strong lenses. Color and more importantly shape information from overlapping space observations will allow for significantly improved constraints on cosmology via enhanced control of lens profile modeling uncertainties.

5.2 WFIRST wide synergies with CMB surveys

The new generation of cosmic microwave background (CMB) experiments, e.g. the Simons Observatory (SO) and CMB-S4, will be well underway by the mid-2020s. In particular, the survey of the Large Aperture Telescope of the Simons Observatory over the full LSST footprint (Ade et al. 2019) will likely be completed or near completion, providing a detailed map of small-scale secondary CMB anisotropies through CMB lensing (reaching a signal-to-noise ratio above 1 for all multipoles $\ell < 200$), the Sunyaev-Zeldovich' (SZ) effect and the cosmic infrared background. The increased number density of galaxies and precision in shape measurements from space-based imaging from WFIRST over the full LSST area would therefore enable a new level of cross-correlations measurements between galaxy surveys and CMB surveys.

The CMB lensing kernel function,

$$q_{\kappa_{\rm CMB}}(\chi) = \frac{3H_0^2\Omega_m}{2c^2}\frac{\chi}{a(\chi)}\frac{\chi^*-\chi}{\chi^*}\,. \tag{32}$$

peaks at $z \sim 2$ and is sensitive to all large-scale structures between the observer and the last scattering surface (here χ^* is the comoving distance to this surface). Measurements of cross-correlations of galaxy clustering and shapes with CMB lensing would strongly benefit from the higher density of galaxies of a joint catalog, which will in turn be beneficial to constraints on cosmic acceleration in several ways. First, extra cosmological information is held in these cross-correlations, which can alleviate parameters degeneracies, for instance between galaxy clustering biases and cosmological parameters. Second, these measurements will enable further cross-calibration of shear multiplicative biases, as shown in Schaan et al. (2017), and other nuisance parameters such as uncertainties on photometric redshifts. Third, the joint catalog will have higher density of galaxies at high redshifts. Therefore, the high redshift tomographic bins will correlate even more strongly with CMB lensing. At last, the comparison of constraints obtained from subsets of these high-precision cross-correlation measurements will enable various tests of the consistency of the cosmological model with all observables, or reveal potential tensions.

Similarly, thermal SZ observations from CMB surveys will detect a wealth of clusters (16,000 clusters forecasted for SO), all of which benefit strongly from cluster mass calibration through high-precision weak lensing (see e.g. Miyatake et al. 2019; Bocquet et al. 2019, for corresponding applications to the Atacama Cosmology Telescope and the South Pole Telescope). The more accurate photometry of the joint catalog would improve the redshift determination of these clusters and extend the redshift range of clusters for which both CMB and optical/infrared observations are matched. This will also enable improved measurements of cluster profiles and features such as the splashback radius observed in density and lensing profiles.

6 CONCLUSIONS

LSST, SPHEREx, Euclid, WFIRST in combination with spectroscopic information (DESI, PFS, 4MOST) and CMB surveys (Simons Observatory, CMB-S4) pose exciting opportunities for discovery in cosmological physics. WFIRST will likely be the last mission to join this ensemble of experiments, but given its versatile observing capabilities and its flexibility in terms of survey strategy, being last is where WFIRST can synergize best.

In this paper we focus on synergies between the WFIRST (space-based, high resolution imaging, NIR multi-band photometry, grism spectroscopy) and LSST (6-band photometry in the visible wavelengths, 10 deg^2 FoV, rapid and repeated coverage of 18,000 deg^2). We explore alternative scenarios to the currently envisioned survey strategy for WFIRST's HLS in particular we quantify scenarios where WFIRST covers the LSST area in one band. Our survey simulations are based on the WFIRST exposure time calculator and redshift distributions from the CANDELS catalog.

As a first result, we find that already a 5-month WFIRST survey in the W-band can cover the full LSST survey area providing high-resolution imaging for >95% of the LSST Year 10 weak lensing galaxy sample. Given that blending is a potentially limiting issue for LSST cosmology, the concept of a 5-month WFIRST wide survey is an important idea to study further.

If WFIRST were to spend 1.5 years covering the LSST area in the W-band it would be able to provide high-resolution imaging for >99% of the LSST Year 10 weak lensing galaxy sample. For this second scenario we run a full 3x2pt likelihood analysis that includes non-Gaussian covariances, and accounts for observational and astrophysical systematics (shear calibration, lens and source photo-z uncertainties, galaxy bias, intrinsic alignment, baryonic physics). For comparison we run similar 3x2pt likelihood analyses for the standard WFIRST HLS survey and for an LSST Year 10 survey. We find a significant increase in constraining power for the joint LSST+WFIRST wide survey compared to LSST Y10 (FoM$_{\rm Wwide}$= 2.6 FoM$_{\rm LSST}$) and compared to WFIRST HLS (FoM$_{\rm Wwide}$= 4.4 FoM$_{\rm HLS}$).

The assumed survey area in both the 5 month and the 1.5 year WFIRST wide survey is 18,000 deg^2, we however note that the best part of LSST's footprint for extragalactic studies might be smaller and correspondingly the required time WFIRST would spend on a wide band survey would be shorter. Such a wide survey component

could be implemented as part of the nominal HLS survey, or as part of the HLS and other WFIRST survey components (depending on their interest); it could also be conducted as part of an extended mission, which would map nicely onto the LSST Y10 timescale.

By the time WFIRST launches, LSST will have been in survey mode for several years already and will have built up substantial survey depth and area. For cosmology and dark energy science from LSST it is not unlikely that blending of source galaxies will be a limiting systematic for LSST shape and photo-z measurements. It is also possible that the lack of deep training data (spectra) for photo-z, or the lack of multi-band IR coverage will limit LSST photo-z accuracy. If the lack of spectroscopic information is limiting, exploring the idea of extended WFIRST grism observations is interesting. If narrow band IR information is useful, a WFIRST wide survey with the H-band is interesting to study. A WFIRST H-band survey is approximately 3.5 times as slow as a W-band survey but it significantly limits wavelength dependent PSF problems.

Exploring optimal joint science strategies for WFIRST and LSST requires complex calculations and meaningful metrics. This paper has presented a connected infrastructure of survey simulations and sophisticated likelihood analyses that can be used to further explore joint LSST+WFIRST science cases. Obviously, a WFIRST wide data set would impact multiple cosmological probes beyond 3x2pt cosmology and we mention cluster cosmology, voids, and trough cosmology, strong lensing, CMB lensing, and SZ synergies as examples for future consideration.

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